

Double integrals in polar co-ordinates

The double integral in the cartesian form

$$\iint_R f(x, y) dx dy \dots (i)$$

Transformed to polar coordinates (i) becomes

$$\iint_R f(r \cos \theta, r \sin \theta) r dr d\theta$$

Triple integrals

$$\iiint_V f(x, y, z) dv = \int_{x_1}^{x_2} \int_{y_1(x)}^{y_2(x)} \int_{z_2(x,y)}^{z_1(x,y)} f(x, y, z) dz dy dx$$

Dirichlet's integrals

If v is a region by $x \geq 0$, $y \geq 0$ and $z \geq 0$ and $x + y + z \leq 1$ then,

$$\iiint_V x^{\ell-1} y^{m-1} z^{n-1} dx dy dz$$

$$= \frac{\Gamma(\ell) \Gamma(m) \Gamma(n)}{\Gamma(\ell + m + n + 1)}$$

1. Area = $\iint_A dx dy$

2. Volume = $\iiint_V dx dy dz$

Handwritten notes:
 $n = r \sin \theta$
 $y = r \sin \theta$
 $dn = r \cos \theta d\theta$

Exercise - 4

1. Evaluate the following integrals.

(i) $\int_0^3 \int_1^2 xy(x+y) dx dy$

Solution: Let $I = \int_0^3 \int_1^2 xy(x+y) dx dy$

$$= \int_0^3 \left[\int_1^2 (x^2y + xy^2) dx \right] dy$$

$$= \int_0^3 \left[\frac{x^3y}{3} + \frac{x^2y^2}{2} \right]_1^2 dy$$

$$= \int_0^3 \left[\frac{8}{3}y + \frac{4y^2}{2} - \frac{1}{3}y - \frac{1}{2}y^2 \right] dy$$

$$= \int_0^3 \left(\frac{7}{3}y + \frac{3y^2}{2} \right) dy$$

Handwritten notes:
Correct
G

$$\begin{aligned}
 &= \left[\frac{7}{3} \cdot \frac{y^2}{2} + \frac{3y^3}{2 \cdot 3} \right]_0^3 \\
 &= \frac{7}{3} \cdot \frac{9}{2} + \frac{1}{2} \cdot 27 - 0 \\
 &= \frac{63}{6} + \frac{27}{2} \\
 &= \frac{63+81}{6} \\
 &= \frac{144}{6}
 \end{aligned}$$

$\therefore I = 24$ Ans.

(ii) $\int_0^3 \int_0^1 (x^2 + 3y^2) dy dx$

Solution: Let $I = \int_0^3 \left[\int_0^1 (x^2 + 3y^2) dy \right] dx$

$$= \int_0^3 \left[x^2 y + \frac{3y^3}{3} \right]_0^1 dx$$

$$= \int_0^3 (x^2 \cdot 1 + 1^3) dx$$

$$= \int_0^3 (x^2 + 1) dx$$

$$= \left[\frac{x^3}{3} + x \right]_0^3$$

$$= \frac{27}{3} + 3 - 0$$

$$= 9 + 3$$

$\therefore I = 12$ Ans.

(iii) $\int_0^2 \int_0^{\sqrt{2x-x^2}} x dx dy$

Solution: Let $I = \int_0^2 \int_0^{\sqrt{2x-x^2}} x dx dy$

$$= \int_0^2 \left[\int_0^{\sqrt{2x-x^2}} x dy \right] dx$$

$$\begin{aligned}
&= \int_0^2 x \cdot [y]_0^{\sqrt{2x-x^2}} dx \\
&= \int_0^2 x \sqrt{2x-x^2} dx \\
&= -\frac{1}{2} \int_0^2 (2-2x-2) \sqrt{2x-x^2} dx \\
&= -\frac{1}{2} \int_0^2 (2-2x) \sqrt{2x-x^2} dx + \frac{1}{2} \cdot 2 \int_0^2 \sqrt{2x-x^2} dx \\
&= -\frac{1}{2} \int_0^2 (2x-x^2)^{\frac{1}{2}} d(2x-x^2) + \int_0^2 \sqrt{1-1+2x-x^2} dx \\
&= -\frac{1}{2} \left[\frac{(2x-x^2)^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_0^2 + \int_0^2 \sqrt{1-(1-2x+x^2)} dx \\
&= -\frac{1}{2} \cdot \frac{2}{3} (4-4)^{\frac{3}{2}} + \int_0^2 \sqrt{1-(1-x^2)} dx \\
&= 0 + \left[\frac{(1-x) \sqrt{1-(1-x)^2}}{2} + \frac{1}{2} \sin^{-1}(1-x) \right]_0^2 \\
&= 0 + \frac{1}{2} \sin^{-1}(-1) - \frac{1}{2} \sin^{-1}(1) \\
&= \frac{1}{2} \cdot \frac{3\pi}{2} - \frac{1}{2} \cdot \frac{\pi}{2} \\
&= \frac{3\pi - \pi}{4} \\
&= \frac{2\pi}{4}
\end{aligned}$$

$$\therefore I = \frac{\pi}{2} \text{ Ans.}$$

(iv) $\int_0^1 \int_0^{\sqrt{x}} (x^2 + y^2) dx dy$

Solution: Let $I = \int_0^1 \int_0^{\sqrt{x}} (x^2 + y^2) dx dy$

$$\begin{aligned}
&= \int_0^1 \left[\int_0^{\sqrt{x}} (x^2 + y^2) dy \right] dx \\
&= \int_0^1 \left[x^2 \cdot y + \frac{y^3}{3} \right]_0^{\sqrt{x}} dx
\end{aligned}$$

$$= \int_0^1 \left[x^2 \cdot x^{\frac{1}{2}} + \frac{\left(x^{\frac{1}{2}}\right)^3}{3} - x^2 \cdot x - \frac{x^3}{3} \right] dx$$

$$= \int_0^1 \left(x^{\frac{5}{2}} + \frac{x^{\frac{3}{2}}}{3} - x^3 - \frac{x^3}{3} \right) dx$$

$$= \int_0^1 \left(x^{\frac{5}{2}} + \frac{x^{\frac{3}{2}}}{3} - \frac{4}{3}x^3 \right) dx$$

$$= \left[\frac{x^{\frac{5}{2}+1}}{\frac{5}{2}+1} + \frac{x^{\frac{3}{2}+1}}{3\left(\frac{3}{2}+1\right)} - \frac{4}{3} \cdot \frac{x^4}{4} \right]_0^1$$

$$= \frac{2}{7} \cdot (1)^{\frac{7}{2}} + \frac{2}{3 \cdot 5} (1)^{\frac{5}{2}} - \frac{1}{3} (1)^4$$

$$= \frac{2}{7} + \frac{2}{15} - \frac{1}{3}$$

$$= \frac{26+30-35}{105}$$

$$= \frac{9}{105}$$

$$\therefore I = \frac{3}{35} \text{ Ans.}$$

(v) $\int_0^1 \int_0^{x^2} e^x dy dx$

Solution: Let $I = \int_0^1 \int_0^{x^2} e^x dy dx$

$$= \int_0^1 \left[\int_0^{x^2} e^x dy \right] dx$$

$$= \int_0^1 \left[\frac{e^x}{\frac{1}{x}} \right]_0^{x^2} dx$$

$$\begin{aligned}
 &= \int_0^1 x \left(e^{\frac{x^2}{x}} - e^0 \right) dx \\
 &= \int_0^1 x (e^x - 1) dx \\
 &= \int_0^1 (xe^x - x) dx \\
 &= \left[x e^x - e^x - \frac{x^2}{2} \right]_0^1 \\
 &= 1 \cdot e^1 - e^1 - \frac{1}{2} + e^0 \\
 &= -\frac{1}{2} + 1 \\
 \therefore I &= \frac{1}{2} \text{ Ans.}
 \end{aligned}$$

(vi) $\int_0^a \int_0^{\sqrt{a^2-y^2}} dx dy$

Solution: Let $I = \int_0^a \left[\int_0^{\sqrt{a^2-y^2}} dx \right] dy$

$$\begin{aligned}
 &= \int_0^a [x]_0^{\sqrt{a^2-y^2}} dy \\
 &= \int_0^a \sqrt{a^2-y^2} dy \\
 &= \left[\frac{y\sqrt{a^2-y^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{y}{a} \right]_0^a \\
 &= 0 + \frac{a^2}{2} \sin^{-1}(1) - \frac{a^2}{2} \sin^{-1}(0) \\
 &= \frac{a^2}{2} \cdot \frac{\pi}{2} - 0 \\
 \therefore I &= \frac{\pi a^2}{4} \text{ Ans.}
 \end{aligned}$$

(vii) $\int_0^1 \int_0^x e^x dy dx$

Solution: Let $I = \int_0^1 \left[\int_0^x e^x dy \right] dx$

$$\begin{aligned}
 &= \int_0^1 \left[\frac{e^x}{\frac{1}{x}} \right]_0^x dx \\
 &= \int_0^1 x (e^1 - e^0) dx \\
 &= \int_0^1 x (e^1 - 1) dx \\
 &= (e - 1) \int_0^1 x dx \\
 &= (e - 1) \left[\frac{x^2}{2} \right]_0^1
 \end{aligned}$$

$$\therefore I = \frac{1}{2}(e - 1) \text{ Ans.}$$

(viii) $\int_1^{\log 8} \int_0^{\log y} e^{x+y} dx dy$

Solution : Let $I = \int_1^{\log 8} \int_0^{\log y} e^{x+y} dx dy$

$$\begin{aligned}
 &= \int_1^{\log 8} \left[\int_0^{\log y} e^x \cdot e^y dx \right] dy \\
 &= \int_1^{\log 8} e^y \left[e^x \right]_0^{\log y} dy \\
 &= \int_1^{\log 8} e^y \left[e^{\log y} - e^0 \right] dy \\
 &= \int_1^{\log 8} (y^{e^y} - e^y) dy \\
 &= \left[y e^y - e^y - e^y \right]_1^{\log 8} \\
 &= \log^8 e^{\log 8} - e^{\log 8} - e^{\log 8} - 1 \cdot e^1 + e^1 + e^1 \\
 &= 8 \log 8 - 8 - 8 + e \\
 \therefore I &= 8 \log 8 - 16 + e \text{ Ans.}
 \end{aligned}$$

(ix) $\int_0^1 \int_y^{\sqrt{y}} (x^2 + y^2) dy dx$

Solution: Let $I = \int_0^1 \int_y^{\sqrt{y}} (x^2 + y^2) dy dx$

$$= \int_0^1 \left[\int_y^{\sqrt{y}} (x^2 + y^2) dx \right] dy$$

$$= \int_0^1 \left[\frac{x^3}{3} + y^2 x \right]_y^{\sqrt{y}} dy$$

$$= \int_0^1 \left(\frac{y^{\frac{3}{2}}}{3} + y^{\frac{5}{2}} - \frac{y^3}{3} - y^3 \right) dy$$

$$= \int_0^1 \left(\frac{y^{\frac{3}{2}}}{3} + y^{\frac{5}{2}} - \frac{4y^3}{3} \right) dy$$

$$= \left[\frac{y^{\frac{3}{2}+1}}{3\left(\frac{3}{2}+1\right)} + \frac{y^{\frac{5}{2}+1}}{\frac{5}{2}+1} - \frac{4y^4}{3 \cdot 4} \right]_0^1$$

$$= \frac{2(1)^{\frac{5}{2}}}{15} + \frac{2(1)^{\frac{7}{2}}}{7} - \frac{1}{3}(1)^4$$

$$= \frac{2}{15} + \frac{2}{7} - \frac{1}{3}$$

$$= \frac{14+30-35}{105}$$

$$= \frac{9}{105}$$

$$\therefore I = \frac{3}{35} \text{ Ans.}$$

(x) $\int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{a^2-x^2-y^2} \, dy \, dx$

Solution: Let $I = \int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{a^2-x^2-y^2} \, dy \, dx$

$$= \int_0^a \left[\int_0^{\sqrt{a^2-y^2}} \left\{ \sqrt{(a^2-y^2)^2 - x^2} \right\} dx \right] dy$$

$$= \int_0^a \left[\frac{x \sqrt{a^2-y^2-x^2}}{2} + \frac{\left(\sqrt{a^2-y^2} \right)^2}{2} \sin^{-1} \frac{x}{\sqrt{a^2-y^2}} \right]_0^{\sqrt{a^2-y^2}} dy$$

$$= \int_0^a \left(\frac{a^2 - y^2}{2} \right) \cdot \sin^{-1}(1) dy$$

$$= \frac{1}{2} \int_0^a (a^2 - y^2) \frac{\pi}{2} dy$$

$$= \frac{\pi}{4} \left[a^2 y - \frac{y^3}{3} \right]_0^a$$

$$= \frac{\pi}{4} \left(a^2 a - \frac{a^3}{3} \right)$$

$$= \frac{\pi}{4} \frac{(3a^3 - a^3)}{3}$$

$$= \frac{\pi}{4} \cdot \frac{2a^3}{3}$$

$$\therefore I = \frac{\pi a^3}{6} \text{ Ans.}$$

2. Evaluate $\iint (x + y)^2 dx dy$ over the area bounded by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Solution: Let $I = \iint (x + y)^2 dx dy$

Here the area of integration A is bounded by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \dots\dots\dots (i)$$

When $y = 0$ in (i) we get,

$$x = \pm a$$

$$\text{and } \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

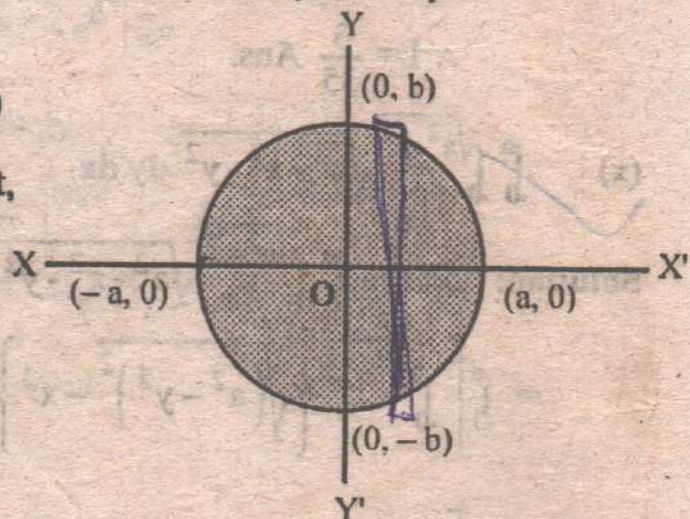
$$\therefore y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

Thus,

$$I = \int_A (x + y)^2 dx dy$$

$$= \int_{x=-a}^{x=a} \left[\int_{y=-\frac{b}{a}\sqrt{a^2-x^2}}^{y=\frac{b}{a}\sqrt{a^2-x^2}} (x^2 + 2xy + y^2) dy \right] dx$$

$$= \int_{-a}^a \left[2 \int_0^{\frac{b}{a}\sqrt{a^2-x^2}} (x^2 + y^2) dy \right] dx$$



Exercise - 6

1. Evaluate $\int_0^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dx dy dz$.

Solution:

$$\begin{aligned}
 \text{Let } I &= \int_0^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dx dy dz \\
 &= \int_0^1 \int_0^z \left[\int_{x-z}^{x+z} (x+y+z) dy \right] dx dz \quad \checkmark \\
 &= \int_0^1 \int_0^z \left[xy + \frac{y^2}{2} + zy \right]_{x-z}^{x+z} dx dz \\
 &= \int_0^1 \int_0^z x(x+z-x+z) + \frac{(x+z)^2 - (x-z)^2}{2} + z(x+z-x+z) dx dz \\
 &= \int_0^1 \int_0^z [2xz + 2xz + 2z^2] dx dz \\
 &= \int_0^1 \int_0^z (4xz + 2z^2) dx dz \\
 &= \int_0^1 \left[\frac{4x^2z}{2} + 2z^2 \cdot x \right]_0^z dz \\
 &= \int_0^1 (2z^3 + 2z^3) dz \\
 &= \int_0^1 4z^3 dz \\
 &= 4 \left[\frac{z^4}{4} \right]_0^1 = \frac{4}{4} [1^4 - 0^4] = 1
 \end{aligned}$$

$$\therefore I = 1.$$

2. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz dz dy dx$

Solution:

$$\begin{aligned}
 \text{Let } I &= \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz dz dy dx \\
 &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\int_0^{\sqrt{1-x^2-y^2}} xyz dz \right] dy dx
 \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[xy \frac{z^2}{2} \right]_0^{\sqrt{1-x^2-y^2}} dy dx \\
&= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} xy(1-x^2-y^2) dy dx. \\
&= \frac{1}{2} \int_0^1 \left[\int_0^{\sqrt{1-x^2}} (xy - x^3y - xy^3) dy \right] dx. \\
&= \frac{1}{2} \int_0^1 \left[\frac{xy^2}{2} - \frac{x^3y^2}{2} - \frac{xy^4}{4} \right]_0^{\sqrt{1-x^2}} dx. \\
&= \frac{1}{2} \int_0^1 \left[\frac{x(1-x^2)}{2} - \frac{x^3(1-x^2)}{2} - \frac{x(1-x^2)^2}{4} \right] dx. \\
&= \frac{1}{2} \int_0^1 \left[\frac{2x - 2x^3 - 2x^3 + 2x^5 - x + 2x^3 - x^5}{4} \right] dx. \\
&= \frac{1}{8} \int_0^1 (x - 2x^3 + x^5) dx. \\
&= \frac{1}{8} \left[\frac{x^2}{2} - \frac{2x^4}{4} + \frac{x^6}{6} \right]_0^1 \\
&= \frac{1}{8} \left(\frac{1}{2} - \frac{2}{4} + \frac{1}{6} \right) \\
&= \frac{1}{8} \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right) \\
&= \frac{1}{48}
\end{aligned}$$

$$\therefore I = \frac{1}{48}$$

3. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{x-x-y} \frac{1}{(x+y+z+1)^3} dx dy dz$

Solution:

$$\begin{aligned}
\text{Let } I &= \int_0^1 \int_0^{1-x} \int_0^{x-x-y} \frac{1}{(x+y+z+1)^3} dx dy dz \\
&= \int_0^1 \int_0^{1-x} \left[\int_0^{x-x-y} (x+y+z+1)^3 dz \right] dy dx.
\end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \int_0^{1-x} \left[\frac{(x+y+z+1)^{-2}}{-2} \right]_0^{1-x-y} dy dx \\
&= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left[\frac{1}{(x+y+z+1)^2} \right]_0^{1-x-y} dy dx \\
&= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left[\frac{1}{(x+y+1-x-y+1)^2} - \frac{1}{(x+y+1)^2} \right] dy dx \\
&= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left[\frac{1}{4} - \frac{1}{(x+y+1)^2} \right] dy dx \\
&= -\frac{1}{2} \int_0^1 \left[\frac{y}{4} - \frac{(x+y+1)^{-1}}{-1} \right]_0^{1-x} dx \\
&= -\frac{1}{2} \int_0^1 \left[\frac{y}{4} + \frac{1}{(x+y+1)} \right]_0^{1-x} dx \\
&= -\frac{1}{2} \int_0^1 \left[\frac{1-x}{4} + \frac{1}{(x+1-x+1)} - \frac{1}{x+1} \right] dx \\
&= -\frac{1}{2} \int_0^1 \left(\frac{1-x}{4} + \frac{1}{2} - \frac{1}{x+1} \right) dx \\
&= -\frac{1}{2} \left[\frac{(1-x)^2}{4 \cdot 2} + \frac{1}{2} x - \log(x+1) \right]_0^1 \\
&= -\frac{1}{2} \left\{ \left(0 + \frac{1}{8}\right) + \frac{1}{2}(1-0) - \log 2 + \log 1 \right\} \\
&= -\frac{1}{2} \left(\frac{1}{8} + \frac{1}{2} - \log 2 \right) \\
&= -\frac{1}{2} \left(\frac{5}{8} - \log 2 \right) \\
&= \frac{1}{2} \left(\log 2 - \frac{5}{8} \right) \\
\therefore I &= \frac{1}{2} \left(\log 2 - \frac{5}{8} \right).
\end{aligned}$$

4. Evaluate $\int_0^4 \int_0^{2\sqrt{z}} \int_0^{\sqrt{4z-x^2}} dz dx dy$

Solution:

$$\text{Let } I = \int_0^4 \int_0^{2\sqrt{z}} \int_0^{\sqrt{4z-x^2}} dz dx dy$$

$$\begin{aligned}
&= \int_0^4 \int_0^{2\sqrt{z}} \left[\int_0^{\sqrt{4z-x^2}} dy \right] dx dz \\
&= \int_0^4 \int_0^{2\sqrt{z}} [y]_0^{\sqrt{4z-x^2}} dx dz \\
&= \int_0^4 \int_0^{2\sqrt{z}} (4z-x^2)^{\frac{1}{2}} dx dz \\
&= \int_0^4 \left[\int_0^{2\sqrt{z}} \sqrt{(2\sqrt{z})^2 - x^2} dx \right] dz \\
&= \int_0^4 \left[\frac{x\sqrt{4z-x^2}}{2} - \frac{4z}{2} \sin^{-1} \frac{x}{2\sqrt{z}} \right]_0^{2\sqrt{z}} dz \\
&= \int_0^4 \left[0 - 0 - \frac{4z}{2} \{ \sin^{-1}(1) - \sin^{-1} 0 \} \right] dz \\
&= \int_0^4 2z \sin^{-1}(1) dz \\
&= \int_0^4 2z \cdot \frac{\pi}{2} dz \\
&= \pi \int_0^4 z dz \\
&= \pi \left[\frac{z^2}{2} \right]_0^4 \\
&= \pi \left[\frac{4^2}{2} \right] \\
&= \pi \cdot \frac{16}{2} \\
&= 8\pi
\end{aligned}$$

$$\therefore I = 8\pi.$$

5. Evaluate $\iiint_V (x-2y+z) dx dy dz$ where V is the region bounded by $0 \leq x \leq 1, 0 \leq y \leq x^2, 0 \leq z \leq x+y$.

Solution:

We have, $\iiint_V (x-2y+z) dx dy dz$ bounded by $0 \leq x \leq 1, 0 \leq y \leq x^2, 0 \leq z \leq x+y$.

$$\begin{aligned}
 \text{Let } I &= \int_0^1 \int_0^{x^2} \left[\int_0^{x+y} (x-2y+z) dz \right] dy dx \\
 &= \int_0^1 \int_0^{x^2} \left[xz - 2yz + \frac{z^2}{2} \right]_0^{x+y} dy dx. \\
 &= \int_0^1 \int_0^{x^2} \left[x(x+y) - 2y(x+y) + \frac{(x+y)^2}{2} \right] dy dx. \\
 &= \int_0^1 \int_0^{x^2} \left[x^2 + xy - 2xy - 2y^2 + \frac{x^2 + 2xy + y^2}{2} \right] dy dx \\
 &= \int_0^1 \int_0^{x^2} \left[x^2 - xy - 2y^2 + \frac{x^2 + 2xy + y^2}{2} \right] dy dx \\
 &= \frac{1}{2} \int_0^1 \int_0^{x^2} (2x^2 - 2xy - 4y^2 + x^2 + 2xy + y^2) dy dx. \\
 &= \frac{1}{2} \int_0^1 \int_0^{x^2} (3x^2 - 3y^2) dy dx. \\
 &= \frac{1}{2} \int_0^1 \left[3x^2 y - \frac{3y^3}{3} \right]_0^{x^2} dx. \\
 &= \frac{1}{2} \int_0^1 [3x^2 \cdot x^2 - (x^2)^3] dx. \\
 &= \frac{1}{2} \int_0^1 (3x^4 - x^6) dx. \\
 &= \frac{1}{2} \left[\frac{3x^5}{5} - \frac{x^7}{7} \right]_0^1 \\
 &= \frac{1}{2} \left(\frac{3}{5} - \frac{1}{7} \right) \\
 &= \frac{1}{2} \left(\frac{21-5}{35} \right) = \frac{1}{2} \cdot \frac{16}{35} = \frac{8}{35} \\
 \therefore I &= \frac{8}{35}.
 \end{aligned}$$

6. Evaluate $\iiint_V x^2 dx dy dz$ over the region V bounded by the planes $x=0$, $y=0$, $z=0$ and $x+y+z=a$

Solution:

We have, $\iiint_V x^2 dx dy dz$
 bounded by the planes $x=0$, $y=0$, $z=0$ and $x+y+z=a$

$$\text{Let } I = \iiint_V x^2 dx dy dz$$

$$= \int_0^a \int_0^{a-x} \int_0^{a-x-y} x^2 dy dz dx$$

$$= \int_0^a \int_0^{a-x} \left[x^2 z \right]_0^{a-x-y} dy dx.$$

$$= \int_0^a \int_0^{a-x} x^2 (a-x-y) dy dx.$$

$$= \int_0^a \int_0^{a-x} (ax^2 - x^3 - x^2 y) dy dx$$

$$= \int_0^a \left[ax^2 y - x^3 y - \frac{x^2 y^2}{2} \right]_0^{a-x} dy dx.$$

$$= \int_0^a \left[ax^2(a-x) - x^3(a-x) - x^2 \frac{(a-x)^2}{2} \right] dx.$$

$$= \int_0^a \left[a^2 x^2 - ax^3 - ax^3 + x^4 - x^2 \frac{(a^2 - 2ax + x^2)}{2} \right] dx.$$

$$= \frac{1}{2} \int_0^a \left[2a^2 x^2 - 4ax^3 + 2x^4 - x^2 a^2 + 2ax^3 - x^4 \right] dx.$$

$$= \frac{1}{2} \int_0^a \left[a^2 x^2 - 2ax^3 + x^4 \right] dx.$$

$$= \frac{1}{2} \left[\frac{a^2 x^3}{3} - \frac{2ax^4}{2} + \frac{x^5}{5} \right]_0^a$$

$$= \frac{1}{2} \left(a^2 \frac{a^3}{3} - \frac{aa^4}{2} + \frac{a^5}{5} \right)$$

$$= \frac{1}{2} \left(\frac{a^5}{3} - \frac{a^5}{2} + \frac{a^5}{5} \right)$$

$$= \frac{1}{2} \left[\frac{10a^5 - 15a^5 + 6a^5}{30} \right]$$

$$= \frac{1}{2} \cdot \frac{a^5}{30}$$

$$= \frac{a^5}{60}$$

$$\therefore I = \frac{a^5}{60}.$$